

Tensors

Tensor is basically a **data structure** used to store data in Machine Learning and Deep Learning.

Most popular ML/DL libraries like:

- TensorFlow
- scikit-learn
- PyTorch

use tensors internally for calculations.

Tensor and NumPy Array

Tensor and NumPy ndarray are very similar because both are used to store numerical data in multiple dimensions.

Main difference:

- NumPy arrays are mainly used for numerical computing.
 - Tensors are optimized for Deep Learning and can run on GPUs.
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Important Terms

Rank / Dimension / Number of Axes

These terms represent how many axes a tensor has.

- 0D tensor → 0 axis
- 1D tensor → 1 axis
- 2D tensor → 2 axes
- ND tensor → N axes

So,

Number of axes = Rank = Dimension of tensor

Shape of Tensor

Shape tells the size of the tensor.

Example:

```
[[1,2,3],[4,5,6]][[1,2,3], [4,5,6]][[1,2,3],[4,5,6]]
```

Shape = $2 \times 3 \times 3$

because:

- 2 rows
 - 3 columns
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Types of Tensors

0D Tensor (Scalar)

A tensor containing only a single value.

Example:

```
2, 3, 102, \ 3, \ 102, 3, 10
```

- No axis
 - Rank = 0
 - Called **Scalar**
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1D Tensor (Vector)

Example:

```
[1,2,3,4][1,2,3,4][1,2,3,4]
```

- One axis
- Rank = 1
- Also called **Vector**

Shape:

(4,)(4,)(4,)

because it contains 4 elements.

2D Tensor (Matrix)

Example:

[[1,2,3],[4,5,6]][[1,2,3], [4,5,6]][[1,2,3],[4,5,6]]

- Two axes
- Rank = 2
- Also called **Matrix**
- Collection of vectors

Shape:

(2,3)(2,3)(2,3)

because:

- 2 rows
 - 3 columns
-

ND Tensor (Higher Dimensional Tensor)

Tensor having more than 2 dimensions.

Example of 3D tensor:

[[[1,2],[3,4]],[[5,6],[7,8]]][[[1,2],[3,4]], [[5,6],[7,8]]] [[[1,2],[3,4]],[[5,6],[7,8]]]

Shape:

(2,2,2)(2,2,2)(2,2,2)

These tensors are heavily used in:

- Image Processing
 - Deep Learning
 - Video Processing
 - NLP
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Real Life Example

- Scalar → Temperature value
 - Vector → List of student marks
 - Matrix → Grayscale image
 - 3D Tensor → Color image (Height × Width × Channels)
 - 4D Tensor → Batch of images in Deep Learning
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Important Interview Point

In Deep Learning:

- Data is mostly stored as tensors.
- Neural Networks perform tensor operations like:
 - addition
 - multiplication
 - reshaping
 - matrix operations

That is why tensors are very important in ML and AI.